

version here is Ubuntu 16.04.

```
$ echo "https://apt.dockerproject.org/
repo ubuntu-xenial main"
```

```
| sudo tee /etc/apt/sources.list.d/
docker.list
```

Step 5: Now update the packages on the Ubuntu system by using the following command:

```
$ sudo apt-get update
```

To verify whether the package manager is pointing to the right repository, run the following command:

```
$ sudo apt-cache policy docker-engine
```

Step 6: You have to install the *linux-image-extra-** kernel packages, which allow one to use the *aufs* storage driver (this driver is used by the newer versions of Docker). This can be done by using the following command:

```
$ sudo apt-get install linux-image-
extra-$(uname -r)
linux-image-extra-virtual
```

Step 7: The final step is to install Docker, as follows:

```
$ sudo apt-get install -y docker-engine
```

Here, *apt-get* uses the *install* option to download the Docker-engine image from the Docker website and get Docker installed. The Docker-engine is the official package from the Docker Corporation for Ubuntu based systems.

To check the Docker version running currently, use the following command:

```
$ sudo docker version
```

To get detailed information on the Docker service installed, enter the

following command:

```
$ sudo docker info
```

This will give information on:

- The number of containers
- The number of images
- The storage driver used by Docker
- The root directory used by Docker
- The execution driver used by Docker

With this, Docker is installed in your machine. To understand more about its functionalities, you need to learn about the Docker Hub.

Docker Hub

Docker Hub is the world's largest public repository of container images with an array of content sources including container community developers, open source projects and independent software vendors (ISV) building and distributing their code in containers. Developers can store and share their personal projects and see what the container community is building. Docker Hub is a registry service on the cloud that allows you to download Docker images. To use Docker Hub, go to <https://hub.docker.com/>.

Docker images

An image is an inert, immutable file that's essentially a snapshot of a container. Images are also known as a combination of a file system and of parameters. Images are created with the *build* command and produce a container when started with the command *run*. They are stored in a Docker registry such as *registry.hub.docker.com*.

Local images can be listed by running the following command:

```
$ docker images
```

Let's look at the example of the following command in Docker:

```
$ docker run hello-world
```

In the above case, the Docker command is specific and tells the Docker program on the operating system that something needs to be done.

The *run* command is to mention that we want to create an instance of an image, which is then called a container.

Finally, 'hello-world' represents the image from which the container is made.

The Docker images on the system can be removed via the *docker rmi* command.

```
$ docker rmi ImageID
```

Docker containers

A Docker container is an executable package of software that includes everything needed to run it. An application requires code, runtime, system tools, system libraries and settings. Docker containers are the new form of virtualisation.

To use a programming metaphor, if an image is a class, then a container is an instance of a class, which is also known as the object. Containers are lightweight and portable encapsulations of an environment in which applications can be run.

The running of containers is managed with the Docker *run* command. To run a container in an interactive mode, first launch the Docker container as follows:

```
$ sudo docker run -it hello-world
```

Docker daemon

The background service running on the host that manages building, running and distributing Docker containers is the Docker daemon. The daemon is a process that runs in the operating system to which clients talk. It is what makes the Docker engine work.

To view local running containers, use the following command:

```
$ docker ps -a
```